
Cylindrical Flow Matching for Complex-Valued Medical Image Synthesis

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Abstract

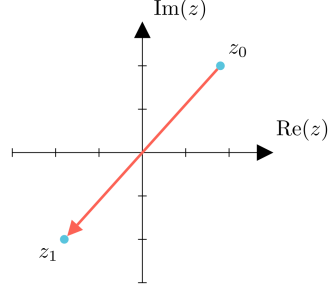
1 While deep generative models have achieved remarkable success in MRI recon-
2 struction, most operate strictly on magnitude images, discarding critical phase
3 information. This magnitude-only paradigm hinders the development and valida-
4 tion of complex-valued neural networks for phase-sensitive diagnostic imaging.
5 Current methods attempting to synthesize fully complex data suffer from a funda-
6 mental topological mismatch: they naively embed the complex signal into a flat
7 Euclidean space ($\mathbb{R}^2$). This ignores the intrinsic circular topology of the phase
8 (S^1), leading to destructive interference, magnitude attenuation, and non-physical
9 artifacts. To address this, we propose **Cylindrical Flow Matching**, a geometry-
10 aware generative framework that explicitly models amplitude on $\mathbb{R}^+$ and phase on
11 S^1 . By constructing probability paths along geodesics on a cylindrical manifold,
12 our approach natively resolves phase discontinuities while preserving the compu-
13 tational efficiency of Optimal Transport. We demonstrate that Cylindrical Flow
14 Matching ensures topological validity and physical signal plausibility, providing a
15 reliable synthetic baseline for safety-critical medical downstream tasks.

16 1 Introduction and Problem Formulation

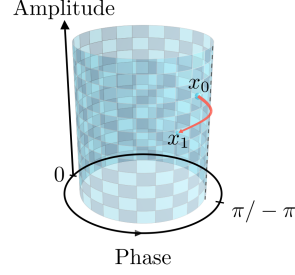
17 Magnetic Resonance Imaging (MRI) is a cornerstone of non-invasive medical diagnostics, yet its
18 inherently slow acquisition time remains a critical bottleneck. Accelerating MRI typically involves
19 undersampling the k -space and solving a highly ill-posed inverse problem [Chung and Ye, 2022].
20 While deep generative models have achieved remarkable success in MRI reconstruction, the vast
21 majority of these approaches operate strictly on magnitude images in the spatial domain, entirely
22 discarding the complex phase information.

23 This magnitude-only paradigm creates a research gap, as emerging complex-valued networks demon-
24 strate superior signal fidelity in raw MRI processing [Rempe et al., 2026, Dou et al., 2025]. However,
25 the lack of realistic, fully complex synthetic data severely limits data augmentation and the robust vali-
26 dation of complex-valued reconstruction algorithms. While recent diffusion approaches [Rempe et al.,
27 2025] explore complex-valued generation, their lack of explicit phase topology preservation leaves
28 a critical need for geometrically grounded baselines. Naive generative models operating without
29 complex-domain geometric constraints are prone to hallucinating non-physical phase discontinuities,
30 which upon Forward Fast Fourier Transform (FFT) leak into measured k -space subspaces and corrupt
31 high-frequency anatomical details.

32 Current approaches suffer from a fundamental topological mismatch: they naively treat the complex
33 signal as a flat, two-channel tensor embedded in a Euclidean space $\mathbb{C} \cong \mathbb{R}^2$ [Chung and Ye, 2022].
34 This ignores the intrinsic circular topology (S^1) of the signal’s phase. Consequently, standard
35 Euclidean interpolation suffers from destructive interference due to the triangle inequality, $|z(t)| =$
36 $|(1-t)z_0 + tz_1| \leq (1-t)|z_0| + t|z_1|$, with strict inequality whenever phases differ ($\theta_0 \neq \theta_1$).
37 Under an independent uniform phase prior $\theta_0 \sim \mathcal{U}(S^1)$, the expected mid-point ($t = 0.5$) magnitude



(a) Standard Euclidean Flow ($\mathbb{R}^2$)



(b) Cylindrical Flow Matching (Ours)

Figure 1: **Topological mismatch in complex-valued generation.** (a) In the complex plane ($\text{Re}(z), \text{Im}(z)$), linear interpolation between antipodal states z_0 and z_1 passes through the origin, causing amplitude attenuation ($|z_{1/2}| \ll \frac{1}{2}(|z_0| + |z_1|)$). (b) Cylindrical Flow Matching follows the shortest angular path on S^1 while interpolating amplitude separately.

attenuation is analytically $1 - \mathbb{E}|\cos(\Delta\theta/2)| = 1 - 2/\pi \approx 36.3\%$ (empirically $\sim 16\%$ on spatially correlated MRI data). Furthermore, flat angular interpolation does not respect wrapping at the boundary ($-\pi \sim \pi$). These effects motivate evaluating a geometry-aware probability path for complex MRI data.

To bridge the gap between generative modeling and the structured nature of physical signals, we propose **Cylindrical Flow Matching**, a geometry-aware generative framework designed specifically for complex-valued medical image synthesis. Building on flow matching [Lipman et al., 2023] and its manifold extension [Chen and Lipman, 2024], we hypothesize that radial-angular paths provide a better inductive bias than Cartesian coordinates, natively resolving phase discontinuities.

In summary, our contributions are as follows:

- **Mathematical Foundations for Learning Dynamics:** We demonstrate how the geometry of the state space shapes probability paths, analytically proving that flat Euclidean paths suffer from non-physical destructive interference ($1 - \mathbb{E}[\cos(\Delta\theta/2)] \approx 36.3\%$ mid-point attenuation), whereas our paths follow intrinsic cylindrical geodesics.
- **Structure-Preserving Dynamics:** We show how continuous trigonometric embeddings and intrinsic velocity regression globally bound angular velocity ($|u_\theta| \leq \pi$), guaranteeing numerical stability for continuous ODE/SDE solvers (e.g., 2nd-order Heun).
- We demonstrate the practical value of this geometry-aware prior in a downstream reconstruction experiment using simulated Cartesian $4 \times k$ -space undersampling.

2 Related Work

Recent advances in **Stochastic Dynamics and Geometry** have demonstrated that tailoring generative models to the intrinsic topology of data significantly improves physical validity. De Bortoli et al. [2022] established the foundations for Riemannian Score-Based Generative Modeling by solving stochastic differential equations (SDEs) on compact manifolds, while De Bortoli et al. [2021] introduced Diffusion Schrödinger Bridges to generalize score-based modeling with theoretically grounded Optimal Transport paths. More recently, Flow Matching [Lipman et al., 2023] provided a simulation-free approach for continuous normalizing flows, which Chen and Lipman [2024] extended to general geometries, enabling scalable training on Riemannian manifolds. Our approach connects these frameworks to phase topology by leveraging closed-form Optimal Transport on the circle [Delon et al., 2010], establishing a structure-preserving dynamical system specifically tailored for complex-valued inverse problems.

3 Methodology: Cylindrical Flow Matching

To construct a topology-aware generative model for complex-valued MRI data, we formulate continuous normalizing flows directly on the intrinsic manifold of the signal [Lipman et al., 2023, Chen

and Lipman, 2024]. Each complex pixel $z = Ae^{i\theta} \in \mathbb{C}$ naturally resides on a cylindrical manifold $\mathcal{M} = \mathbb{R}^+ \times S^1$, equipped with the product metric $g = dA^2 + d\theta^2$, such that a 2D complex image is modeled on $\mathcal{M}^{H \times W}$. In contrast, the Euclidean baseline naively operates on 2-channel Cartesian inputs $(\text{Re}(z), \text{Im}(z)) \in \mathbb{R}^2$, regressing straight Euclidean paths $z_t = (1-t)z_0 + tz_1$ toward target velocity $u_{\text{euc}} = z_1 - z_0$ with an unweighted L_1 objective $\mathcal{L}_{\text{euc}}(\phi) = \mathbb{E}[|v_x - u_x| + |v_y - u_y|]$. Our cylindrical formulation explicitly decouples amplitude and phase transport: geodesics interpolate each component independently along the shortest angular path on S^1 , preventing the amplitude collapse inherent to Euclidean interpolation where antipodal trajectories ($\Delta\theta = \pi$) are forced through the origin ($A \rightarrow 0$). Furthermore, this avoids the singular polar metric $g_{\text{polar}} = dA^2 + A^2 d\theta^2 \cong \mathbb{C} \setminus \{0\}$, whose straight Euclidean geodesics also collapse near the removed origin.

While the cylinder possesses vanishing Riemann curvature ($K = 0$), it exhibits a non-trivial fundamental group ($\pi_1(\mathcal{M}) = \mathbb{Z}$), which causes phase seam defects in flat models. Moreover, the cylinder is a parallelizable manifold with trivial global frame $(\partial_A, \partial_\theta)$, allowing direct velocity regression without pushforward operators. Crucially, under Cartesian flow, the induced angular velocity $\dot{\theta} = (xu_y - yu_x)/A^2 = \mathcal{O}(A^{-1})$ diverges near the origin $A \rightarrow 0$ with unbounded variance. In contrast, our cylindrical target velocity is globally bounded, $|u_\theta| \leq \pi$, guaranteeing numerical stability. Furthermore, the product metric cost decomposes separably, creating a natural theoretical bridge to closed-form Optimal Transport on S^1 via circular CDF shifts [Delon et al., 2010].

3.1 Continuous Network Embedding

To circumvent artificial jump discontinuities at the branch cut $[-\pi, \pi)$, our network operates on a 3-channel continuous trigonometric embedding $x = (A, \cos \theta, \sin \theta) \in \mathbb{R}^3$ as input. The neural network then predicts a 2-channel velocity field $u_t = (v_A, v_\theta) \in T_{(A,\theta)}\mathcal{M}$, corresponding to the tangent space of the amplitude and phase.

3.2 Loss Formulation

Given paired samples $x_0 = (A_0, \theta_0) \sim p_0$ and $x_1 = (A_1, \theta_1) \sim p_1$ (where p_1 denotes the empirical target distribution of complex-valued MRI data), we construct geodesic probability paths along each component: $A_t = (1-t)A_0 + tA_1$ and $\theta_t = \exp_{\theta_0}^{S^1}(t u_\theta)$. The target velocity field on the product manifold is:

$$u_A = A_1 - A_0, \quad u_\theta = \text{atan2}(\sin(\theta_1 - \theta_0), \cos(\theta_1 - \theta_0)) \quad (1)$$

where u_θ computes the shortest signed angular difference via the S^1 logarithmic map, correctly resolving transport across the $\pm\pi$ boundary.

The network $v_\phi(t, x)$ regresses the target velocity field $u_t(x_t)$ via an edge-preserving L_1 objective:

$$\mathcal{L}(\phi) = \mathbb{E}_{t, x_0, x_1} \left[|v_A - u_A| + \lambda \frac{A_1}{\bar{A}_1} |v_\theta - u_\theta| \right] \quad (2)$$

where $\lambda = 1.0$ and $\bar{A}_1$ is the batch mean amplitude. Modulating the phase loss by A_1 acts as an empirical SNR heuristic inspired by phase Fisher information (which formally scales as A_1^2 under complex Gaussian noise), concentrating capacity on tissue while mitigating ill-defined phase in background noise. Furthermore, we model amplitude A linearly rather than logarithmically to avoid numerical instabilities as $A \rightarrow 0$.

3.3 Base Distribution and Inference

We construct our generative process from the base distribution $p_0 = \mathcal{U}([0, 1]) \otimes \mathcal{U}(S^1)$, a uniform prior on the unit cylinder (clamped with $\epsilon = 10^{-6}$ for numerical stability at the origin). Crucially, because the target velocity field is computed via $\mathcal{O}(1)$ closed-form geodesics, training completely bypasses forward simulation; numerical integration (via a 2nd-order Heun solver) is strictly reserved for the inference generation phase.

4 Experiments and Results

We evaluate Cylindrical Flow Matching on complex-valued MRI synthesis and downstream reconstruction using the raw k -space SKM-TEA-mini dataset [Desai et al., 2021] (single-coil, first-echo

Table 1: **Quantitative metrics.** Mean $\pm$ standard deviation across all 512 slices of the held-out test volume ($4\times$ Cartesian reconstruction with exact Data Consistency projections).

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$	CPE (rad) $\downarrow$
Euclidean ($\mathbb{R}^2$)	24.13 ± 1.18	0.530 ± 0.06	0.513 ± 0.07
Cylindrical Flow (Ours)	24.47 ± 1.16	0.588 ± 0.04	0.405 ± 0.05

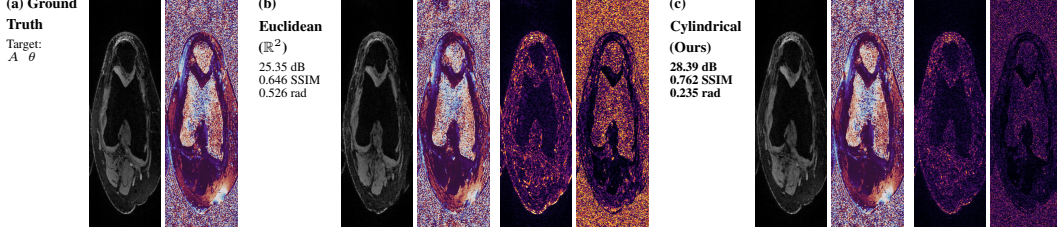


Figure 2: **Qualitative comparison on a held-out test slice.** Columns per method: Reconstructed (A, θ) followed by error maps ($|\Delta A|, |\Delta \theta|$) in Inferno scale. The Euclidean baseline exhibits severe phase seams and boundary magnitude cancellation, natively avoided by Cylindrical Flow.

image space, comprising 3 complete 3D knee volumes with 1536 axial slices total). We partition the dataset into 2 patient volumes for training (1024 slices) and 1 completely held-out patient volume ($N = 512$ contiguous slices) for testing. This constrained data regime provides a rigorous test of topological inductive bias: while Euclidean models must waste parameter capacity trying to approximate non-physical phase branch cuts, our cylindrical formulation natively respects $S^1 \times \mathbb{R}^+$ geometry, yielding superior generalization on unseen anatomy. Retrospective $4\times$ Cartesian undersampling is simulated using 1D phase-encoding masks with a central 24-line autocalibration signal (ACS) band, with exact Data Consistency (DC Error $< 10^{-6}$) enforced via interleaved k -space projections.

To evaluate reconstruction quality, we report standard Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) on magnitude, alongside the Circular Phase Error (CPE) in radians:

$$\text{CPE}(\theta, \hat{\theta}) = \frac{1}{|M|} \sum_{i \in M} |\text{atan2}(\sin(\theta_i - \hat{\theta}_i), \cos(\theta_i - \hat{\theta}_i))| \quad (3)$$

where M denotes a tissue mask thresholded at 5% of peak amplitude to avoid background noise bias.

As presented quantitatively across the full volume in Table 1 (mean $\pm$ std) and qualitatively on a representative central slice in Figure 2, the Euclidean model struggles with phase wrapping, generating artificial branch-cut seams that induce destructive interference and signal cancellations near tissue boundaries. In contrast, by explicitly modeling phase on S^1 , Cylindrical Flow Matching preserves smooth phase transitions and prevents mid-trajectory amplitude collapse, leading to sharper anatomical edges and consistent performance gains across all evaluated metrics.

5 Conclusion and Future Work

By formulating probability paths on $\mathbb{R}^+ \times S^1$, Cylindrical Flow Matching natively resolves phase discontinuities and magnitude attenuation, providing a physically consistent generative baseline for complex-valued MRI. Future work will extend this framework to multi-coil acceleration benchmarks and quantitative phase mapping. Finally, extending this cylindrical manifold formulation to continuous Schrödinger Bridges will enable data-driven priors for downstream complex inverse problems and diagnostic tasks.

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